

Let $\mathbf{q}(t) = \langle h(t), \theta(t) \rangle$ with h the CM vertical displacement and θ the rotation about the flipping axis. The trajectory is piecewise-constant acceleration:

$$\ddot{\mathbf{q}} = \begin{cases} \langle a, \alpha \rangle, & 0 \leq t < t_0 \quad (\text{Forced}) \\ \langle -g, 0 \rangle, & t_0 \leq t \leq t' \quad (\text{Free fall}) \end{cases}$$

with $\mathbf{q}(0) = \dot{\mathbf{q}}(0) = \langle 0, 0 \rangle$ and $\mathbf{q}(t') = \langle 0, n\pi \rangle$.

Q1. Toss Constraint

Phase 1 ($0 \leq t < t_0$): constant acceleration, zero initial conditions. Integrating twice:

$$\dot{\mathbf{q}}(t_0) = t_0 \langle a, \alpha \rangle, \quad \mathbf{q}(t_0) = \frac{1}{2} t_0^2 \langle a, \alpha \rangle.$$

Hence $h_0 \equiv h(t_0) = \frac{1}{2} a t_0^2$, $\dot{h}_0 \equiv \dot{h}(t_0) = a t_0$, $\theta_0 \equiv \theta(t_0) = \frac{1}{2} \alpha t_0^2$, $\dot{\theta}_0 \equiv \dot{\theta}(t_0) = \alpha t_0$.

Phase 2 ($t_0 \leq t \leq t'$): free motion under gravity (no torque). Let $\Delta t = t' - t_0$. Then

$$\begin{aligned} h(t') &= h_0 + \dot{h}_0 \Delta t - \frac{1}{2} g (\Delta t)^2 = \frac{1}{2} a t_0^2 + a t_0 \Delta t - \frac{1}{2} g \Delta t^2, \\ \theta(t') &= \theta_0 + \dot{\theta}_0 \Delta t = \frac{1}{2} \alpha t_0^2 + \alpha t_0 \Delta t. \end{aligned}$$

Success requires $h(t') = 0$ and $\theta(t') = n\pi$. From $h(t') = 0$ (quadratic in Δt , positive root):

$$\Delta t = \frac{t_0}{g} \left(a + \sqrt{a(a+g)} \right). \quad (1)$$

From $\theta(t') = n\pi$:

$$\Delta t = \frac{2\pi n - \alpha t_0^2}{2\alpha t_0}. \quad (2)$$

Equating (1) and (2) gives the **toss constraint**:

$$\boxed{\frac{\alpha t_0^2}{2g} \left(2\sqrt{a(a+g)} + 2a + g \right) = \pi n} \quad \text{or} \quad \boxed{t_0^2 = \frac{2\pi g n}{\alpha \left(2\sqrt{a(a+g)} + 2a + g \right)}}.$$

Q2. Toss Energy Ladder

The pancake is a uniform disk of radius R , mass $m = \pi R^2 \rho$. Its moment of inertia about the flipping axis (in-plane, through CM) is $I = \int_{\Omega} y^2 dm$. In polar coordinates:

$$\begin{aligned} I &= \int_0^{2\pi} \int_0^R (r \sin \theta)^2 \rho r dr d\theta = \rho \int_0^{2\pi} \sin^2 \theta d\theta \int_0^R r^3 dr \\ &= \rho \cdot \pi \cdot \frac{R^4}{4} = \frac{1}{4} m R^2. \end{aligned}$$

Since the axis passes through CM, kinetic energy separates: $E(t) = mgh(t) + \frac{1}{2} m \dot{h}(t)^2 + \frac{1}{2} I \dot{\theta}(t)^2$. At $t = 0$ all terms vanish ($E(0) = 0$). The work done by the pan is $W = E(t_0)$:

$$\begin{aligned} W &= mg \left(\frac{1}{2} a t_0^2 \right) + \frac{1}{2} m (a t_0)^2 + \frac{1}{2} \left(\frac{1}{4} m R^2 \right) (\alpha t_0)^2 \\ &= \frac{m t_0^2}{8} \left(R^2 \alpha^2 + 4a^2 + 4ag \right). \end{aligned}$$

Substituting t_0^2 from the toss constraint ($t_0^2 \propto n$):

$$\boxed{W = \frac{\pi g m n}{4\alpha} \frac{R^2 \alpha^2 + 4a^2 + 4ag}{2\sqrt{a(a+g)} + 2a + g} \propto n}.$$

Q3. Cracked Pancake

The crack splits the disk along the x -axis (the flipping axis) into upper ($y > 0$) and lower ($y < 0$) half-disks, each of mass $m/2$.

Center of mass of a uniform half-disk. By symmetry $\bar{x} = 0$. For $\bar{y}$, integrate over the upper half-disk:

$$\bar{y} = \frac{1}{m/2} \int_{\text{half}} y \, dm = \frac{2}{m} \int_0^\pi \int_0^R (r \sin \theta) \rho r \, dr \, d\theta.$$

With $m = \pi R^2 \rho$:

$$\begin{aligned} \bar{y} &= \frac{2\rho}{\pi R^2 \rho} \int_0^\pi \sin \theta \, d\theta \int_0^R r^2 \, dr = \frac{2}{\pi R^2} \cdot [-\cos \theta]_0^\pi \cdot \frac{R^3}{3} \\ &= \frac{2}{\pi R^2} \cdot 2 \cdot \frac{R^3}{3} = \frac{4R}{3\pi}. \end{aligned}$$

Let $d = \bar{y} = 4R/(3\pi)$ be the CM distance from the flipping axis.

During the forced phase the upper half-disk's CM undergoes circular motion about the x -axis with $\omega(t) = \alpha t$. The centripetal force required is

$$F_{\text{cent}}(t) = \frac{m}{2} \omega(t)^2 d = \frac{2mR}{3\pi} \alpha^2 t^2.$$

The egg yolk supplies this across the crack; the maximum occurs at $t = t_0$: $F_{\text{cent,max}} = \frac{2mR}{3\pi} \alpha^2 t_0^2$.

Using t_0^2 from the toss constraint (Q1):

$$\begin{aligned} F_{\text{cent,max}} &= \frac{2mR}{3\pi} \alpha^2 \frac{2\pi g n}{\alpha(2\sqrt{a(a+g)} + 2a + g)} \\ &= \frac{4R\alpha g m n}{3(2\sqrt{a(a+g)} + 2a + g)}. \end{aligned}$$

Therefore the egg yolk must satisfy

$$F_y \geq \frac{4R\alpha g m n}{3(2\sqrt{a(a+g)} + 2a + g)}.$$

The right-hand side is finite for any finite α , a , n — a finite F_y always suffices. The required force scales linearly with both n and α : more flips or a more aggressive spin demand stronger glue.